

Probability and Statistics

Lecture notes based on the uploaded slides: Lectures 1-4

How to use these notes

The slides are visual and example-driven. These notes turn them into a readable study document: definitions first, then formulas, then worked examples, then checks for interpretation.

The core idea across Lectures 1-4 is: model uncertainty with probability, update uncertainty when information arrives, describe random quantities with distributions, and choose models that match the data-generating situation.

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Lecture 1 - Uncertainty and how to model it

The opening lecture motivates probability and statistics through practical questions: forecasting a volcanic eruption, using project data, pricing a second-hand car, and modelling measurement errors. All of these questions involve uncertainty: the answer is not known with certainty, but we can still reason about it systematically.

1.1 Why probability and statistics are interwoven

Probability starts from a model of randomness and asks what outcomes we should expect. Statistics starts from observed data and asks what model, parameter, or conclusion is plausible. In real work, the two are not separate: we use probability to build statistical tools, and we use statistics to improve probability models.

Perspective	Typical question	Direction
Probability	Given a model, what is the chance of an event?	Model -> consequences
Statistics	Given data, what can we infer about the model?	Data -> model or parameter

1.2 Experiments, sample spaces, outcomes, and events

An experiment is a repeatable procedure whose exact result is uncertain before it happens. The sample space, usually written Ω , is the set of all possible outcomes. An event is a subset of the sample space.

Experiment: roll one fair die

Sample space: $\Omega = \{1,2,3,4,5,6\}$

Outcome: one element of Ω , for example 4

Event: $A = \{2,4,6\}$, the event that the die shows an even number

Interpretation habit

Do not start with formulas. First say what the random experiment is, what the outcome is, and what event you are calculating. This prevents many mistakes in probability problems.

1.3 Set operations for events

Notation	Meaning	Example
$A \cup B$	A or B happens, including both	even or greater than 4
$A \cap B$	A and B both happen	even and greater than 4
A^c	A does not happen	not even
$A \subseteq B$	Whenever A happens, B also happens	rolling 6 implies greater than 4
$A \cap B = \emptyset$	A and B are disjoint; they cannot both happen	rolling 1 and rolling 6

1.4 Probability rules

$$0 \leq P(A) \leq 1$$

$$P(\Omega) = 1$$

If A and B are disjoint: $P(A \cup B) = P(A) + P(B)$

General addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Complement rule: $P(A^c) = 1 - P(A)$

Why subtract $P(A \cap B)$?

When we add $P(A) + P(B)$, outcomes in both A and B have been counted twice. Subtracting $P(A \cap B)$ removes the double count.

1.5 Equal-likelihood models and counting

If all outcomes in a finite sample space are equally likely, probability becomes counting:

$$P(A) = \text{number of outcomes in } A / \text{number of outcomes in } \Omega$$

Example: rolling one fair die. The event $A = \{2,4,6\}$ has 3 favourable outcomes out of 6 total outcomes, so $P(A)=3/6=1/2$.

1.6 Learning workflow expected in the course

The first lecture emphasizes a four-part workflow: preknowledge, formalization, challenge, and verification. In practice this means:

- 1 Ask what you already know and what your intuition says.
- 2 Translate the real-world question into a probability model or statistical question.
- 3 Compute carefully, making each step explicit.
- 4 Interpret the result: what does the number mean, and does it make sense?

Lecture 2 - Conditional probability and Bayes' rule

This lecture focuses on probability when extra information is available. The programme includes conditional probability, the multiplication rule, the law of total probability, Bayes' rule, and independence of events.

2.1 Conditional probability

Conditional probability asks: how likely is A, after we learn that B happened? The event B becomes the new restricted universe.

$$P(A | B) = P(A \cap B) / P(B), \text{ provided } P(B) > 0$$

Read $P(A | B)$ as "the probability of A given B". The denominator $P(B)$ is important: once B is known, probabilities are measured relative to B, not relative to the original whole sample space.

2.2 Multiplication rule

Rearranging the conditional probability definition gives the multiplication rule. It is used when an event is built in stages.

$$P(A \cap B) = P(A | B)P(B)$$

Equivalently: $P(A \cap B) = P(B | A)P(A)$

For several events, multiply probabilities step by step, conditioning on what already happened.

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$$

2.3 Law of total probability

Sometimes an event A can happen through several mutually exclusive cases $B_1, B_2, \dots, B_k$. If these cases partition the sample space, then the probability of A is the sum of the probabilities of A happening through each case.

If $B_1, \dots, B_k$ form a partition of Ω :

$$P(A) = P(A | B_1)P(B_1) + \dots + P(A | B_k)P(B_k)$$

Partition requirement

The cases must be exhaustive and non-overlapping: one of them must happen, and two cannot happen at the same time.

2.4 Bayes' rule

Bayes' rule reverses conditioning. It lets us calculate the probability of a cause or hypothesis after observing evidence.

$$P(B | A) = P(A | B)P(B) / P(A)$$

When A can occur through several cases $B_1, \dots, B_k$, substitute the law of total probability into the denominator:

$$P(B_j | A) = P(A | B_j)P(B_j) / \sum_i P(A | B_i)P(B_i)$$

Term	Bayesian role	Meaning
$P(B)$	Prior	Belief before seeing the evidence

$P(A B)$	Likelihood	How expected the evidence is if B is true
$P(A)$	Evidence / normalizer	Total probability of observing A
$P(B A)$	Posterior	Updated belief after observing A

2.5 Worked example: medical-style test

Suppose 1% of a population has a disease. A test is positive in 95% of diseased people and falsely positive in 10% of healthy people. What is the probability of disease given a positive test?

Let D = disease and $+$ = positive test.

$$P(D)=0.01, P(D^c)=0.99$$

$$P(+|D)=0.95, P(+|D^c)=0.10$$

$$\begin{aligned} P(+) &= P(+|D)P(D) + P(+|D^c)P(D^c) \\ &= 0.95*0.01 + 0.10*0.99 \\ &= 0.0095 + 0.099 \\ &= 0.1085 \end{aligned}$$

$$\begin{aligned} P(D|+) &= P(+|D)P(D) / P(+) \\ &= 0.0095 / 0.1085 \\ &\approx 0.0876 \end{aligned}$$

Interpretation: even with a positive test, the probability of disease is only about 8.8% because the disease is rare and false positives dominate the positive-test group.

2.6 Independence

Events A and B are independent if learning that B happened does not change the probability of A.

A and B independent $\Leftrightarrow P(A | B) = P(A)$, if $P(B)>0$

Equivalent test: $P(A \cap B) = P(A)P(B)$

Common trap

Disjoint does not mean independent. If two non-empty events are disjoint, learning that one happened makes the other impossible, so their probabilities change.

Lecture 3 - Random variables and descriptive statistics

This lecture introduces random variables, density functions, cumulative distribution functions, and descriptive statistics. A random variable turns outcomes into numbers so that uncertainty can be studied quantitatively.

3.1 Random variables

A random variable is a rule that assigns a numerical value to each outcome in the sample space. It is not a single unknown number; it is a function from outcomes to numbers.

$X: \Omega \rightarrow \mathbb{R}$

Example: choose a student at random and let X = birth month.

Then possible values are $1, 2, \dots, 12$.

The birth-month slide uses all TU Delft students as the sample space. Choosing one student at random is the random experiment; asking for the birth month produces the random variable.

3.2 Discrete versus continuous random variables

Type	Values	Probability description	Example
Discrete	Finite or countably many values	Probability mass function $p(x) = P(X=x)$	number of correct answers, birth month
Continuous	Intervals of real numbers	Density function $f(x)$; probabilities are areas	measurement error, height, time

3.3 Probability mass functions and density functions

For a discrete random variable, individual values can have positive probability. For a continuous random variable, individual exact values usually have probability 0; intervals have positive probability.

Discrete: $P(X=x) = p(x)$, and $\sum_x p(x) = 1$

Continuous: $P(a \leq X \leq b) = \int_a^b f(x) dx$, and $\int_{-\infty}^{\infty} f(x) dx = 1$

Density is not probability

For a continuous variable, $f(x)$ can be greater than 1. Probability is area under the density curve, not the height of the curve alone.

3.4 Cumulative distribution function

The cumulative distribution function (CDF) gives the probability that the random variable is at most a given value.

$F(x) = P(X \leq x)$

For continuous variables, the CDF accumulates area under the density. For discrete variables, the CDF is a step function that jumps at possible values.

3.5 Uniform example on [2,5]

The slides use a uniform random variable on $[2,5]$. The density is constant because all intervals of equal length inside $[2,5]$ are equally likely. The interval has length 3, so the density height must be $1/3$.

If $X \sim \text{Uniform}(2,5)$, then
 $f(x) = 1/3$ for $2 \leq x \leq 5$, and 0 otherwise.

$F(x) = 0$ for $x < 2$
 $F(x) = (x-2)/3$ for $2 \leq x \leq 5$
 $F(x) = 1$ for $x > 5$

Therefore $F(3.5) = (3.5-2)/3 = 1.5/3 = 0.5$. Also $F(6.2) = 1$ because 6.2 is above the upper endpoint 5.

3.6 Expected value and variance

The expected value is the long-run average value of a random variable under repeated sampling. Variance measures spread around the expected value.

Discrete: $E[X] = \sum x \cdot p(x)$
Continuous: $E[X] = \int x \cdot f(x) \, dx$
Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$
Standard deviation: $\text{SD}(X) = \sqrt{\text{Var}(X)}$

3.7 Descriptive statistics

Statistic	What it describes	Sensitive to outliers?
Mean	Arithmetic average	Yes
Median	Middle value after sorting	No, more robust
Quantile	Cutoff below which a chosen fraction lies	Usually robust
Variance / standard deviation	Spread around the mean	Yes
Interquartile range	Middle 50% spread: $Q3 - Q1$	More robust

Lecture 4 - Modelling

Lecture 4 moves from definitions to modelling. The programme includes medians and quantiles, families of discrete random variables, families of continuous random variables, and the modelling process.

4.1 Median and quantiles

A median is a value m such that at least half the probability is at or below m and at least half is at or above m . For continuous distributions, it is often defined by $F(m)=0.5$.

Median m : $F(m) = 0.5$, when F is continuous and strictly increasing
p-quantile q_p : $F(q_p) = p$

For $X \sim \text{Uniform}(2,5)$, the median solves $(m-2)/3 = 0.5$, so $m=3.5$.

4.2 Families of discrete random variables

Distribution	When to use it	Parameters	Key formulas
Bernoulli	One yes/no trial	p	$P(X=1)=p$, $E[X]=p$, $\text{Var}(X)=p(1-p)$
Binomial	Number of successes in n independent Bernoulli trials	n, p	$P(X=k)=C(n,k)p^k(1-p)^{(n-k)}$, $E[X]=np$
Geometric	Number of trials until first success	p	$P(X=k)=(1-p)^{(k-1)}p$
Poisson	Counts of rare events in a fixed interval	λ	$P(X=k)=e^{(-\lambda)}\lambda^k/k!$, $E[X]=\lambda$

4.3 Families of continuous random variables

Distribution	When to use it	Parameters	Key idea
Uniform	All values in an interval are equally likely	a, b	Constant density on $[a,b]$
Normal	Symmetric measurement noise or aggregate variation	μ, σ	Bell-shaped; centered at μ
Exponential	Waiting time until the next event in a Poisson process	λ	Memoryless waiting-time model

4.4 Choosing a model

A probability model should match the mechanism that generates the data. Do not choose a distribution only because its formula is familiar. Start with the story of the experiment.

Question to ask	Modelling implication
Is the outcome a count, a category, or a measurement?	Choose discrete for counts/categories, continuous for measurements.

Is there a fixed number of independent trials?	A binomial model may fit.
Are we counting events over time/space?	A Poisson model may fit.
Are we measuring noise around a typical value?	A normal model may fit.
Are all values in an interval equally plausible?	A uniform model may fit.
Are we modelling waiting time until an event?	An exponential model may fit.

4.5 Modelling workflow

- 1 Define the random experiment and the random variable precisely.
- 2 Decide whether the variable is discrete or continuous.
- 3 Choose a distribution family based on the data-generating mechanism.
- 4 Identify the parameters and estimate them from information or data if needed.
- 5 Compute the probability, quantile, expectation, or interval of interest.
- 6 Verify whether the answer is plausible in the original context.

Good modelling is explicit

Write down assumptions such as independence, equal likelihood, fixed number of trials, constant rate, or symmetry. Most modelling errors are hidden assumption errors.

Quick formula sheet

Events and probability

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{If } A \text{ and } B \text{ disjoint: } P(A \cup B) = P(A) + P(B)$$

Conditional probability and Bayes

$$P(A|B) = P(A \cap B) / P(B)$$

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

$$P(B_j|A) = P(A|B_j)P(B_j) / \sum_i P(A|B_i)P(B_i)$$

$$\text{Independence: } P(A \cap B) = P(A)P(B)$$

Random variables

$$\text{CDF: } F(x) = P(X \leq x)$$

$$\text{Discrete expectation: } E[X] = \sum_x x \cdot p(x)$$

$$\text{Continuous expectation: } E[X] = \int x f(x) dx$$

$$\text{Variance: } \text{Var}(X) = E[X^2] - (E[X])^2$$

Common distributions

Name	PMF/PDF or key probability	Mean	Variance
Bernoulli(p)	$P(X=1)=p, P(X=0)=1-p$	p	$p(1-p)$
Binomial(n,p)	$P(X=k) = C(n,k)p^k(1-p)^{n-k}$	np	$np(1-p)$
Poisson(λ)	$P(X=k) = e^{-\lambda} \lambda^k / k!$	λ	λ
Uniform(a,b)	$f(x) = 1/(b-a)$ on $[a,b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$f(x) = \lambda e^{-\lambda x}, x \geq 0$	$1/\lambda$	$1/\lambda^2$

Exam-style problem checklist

Use this checklist when solving problems from these lectures. It forces the formalization and verification steps that the slides emphasize.

- 1 Name the experiment: what random process is happening?
- 2 Define the sample space or the relevant random variable.
- 3 Define events using clear symbols, for example $A = \text{"test is positive"}$.
- 4 Write down the information given in probability notation.
- 5 Choose the rule: addition, complement, multiplication, total probability, Bayes, CDF, expectation, or distribution formula.
- 6 Substitute numbers only after the symbolic formula is correct.
- 7 Compute carefully and keep track of whether the answer is a probability, a density, a count, an expected value, or a quantile.
- 8 Interpret the answer in one sentence using the original context.
- 9 Check plausibility: is the result between 0 and 1 if it is a probability? Does the size make sense? Did conditioning change the denominator?

Mini practice problems

Problem	Core method
A fair die is rolled. What is $P(\text{even or greater than 4})$?	Use addition rule and subtract overlap.
A test has known sensitivity/specificity and disease prevalence. Find $P(\text{disease} \mid \text{positive})$.	Use Bayes' rule and total probability.
$X \sim \text{Uniform}(2,5)$. Find $F(4.1)$.	Use the CDF formula $(x-2)/3$ inside the interval.
Count successes in 20 independent trials with success probability p .	Use $\text{Binomial}(n,p)$.
Count arrivals per hour when the average rate is λ .	Use $\text{Poisson}(\lambda)$.

Source basis: uploaded Probability and Statistics lecture slides for Lectures 1-4: Uncertainty; Conditional probability and Bayes' rule; Random variables and descriptive statistics; Modelling.